QGIS Basics

Lab_01: Mapping cholera in Soho

402.467 / Applied GIS Fundamentals

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14 October 2026

5’ Recap

How GIS applies to everyday questions

Folder structure

GitHub discussions field

File naming best practices in QGIS

Use AI critically!

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What is GIS?

Geographic

Information

System



GIS (noun), is a technology that is used to create, manage, analyze, and map all types of data

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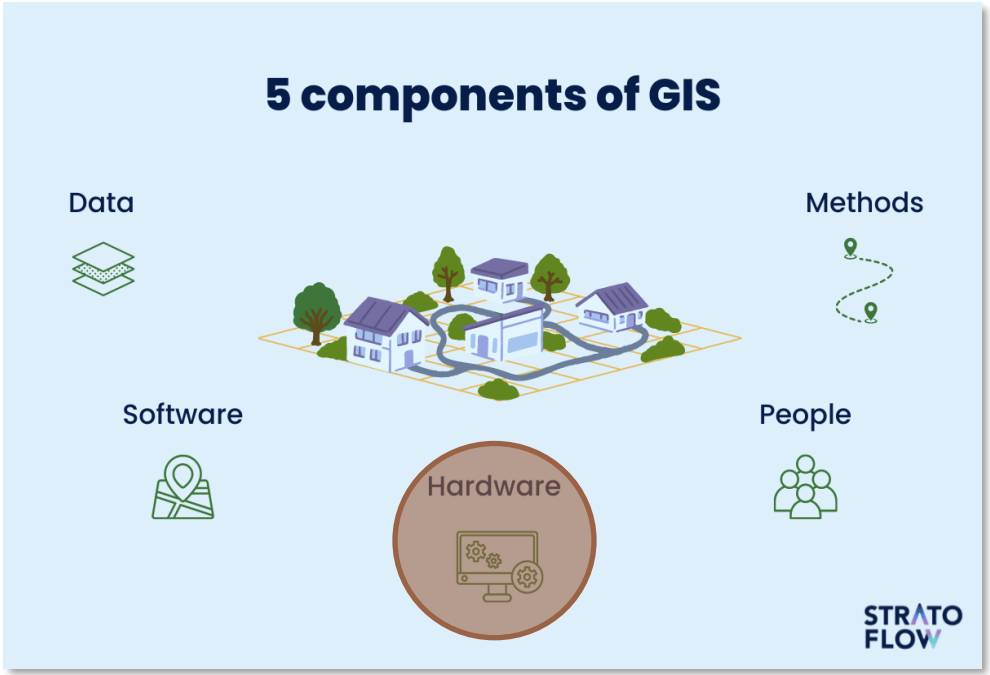


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Key GIS Concept: GIS Components

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GIS Components



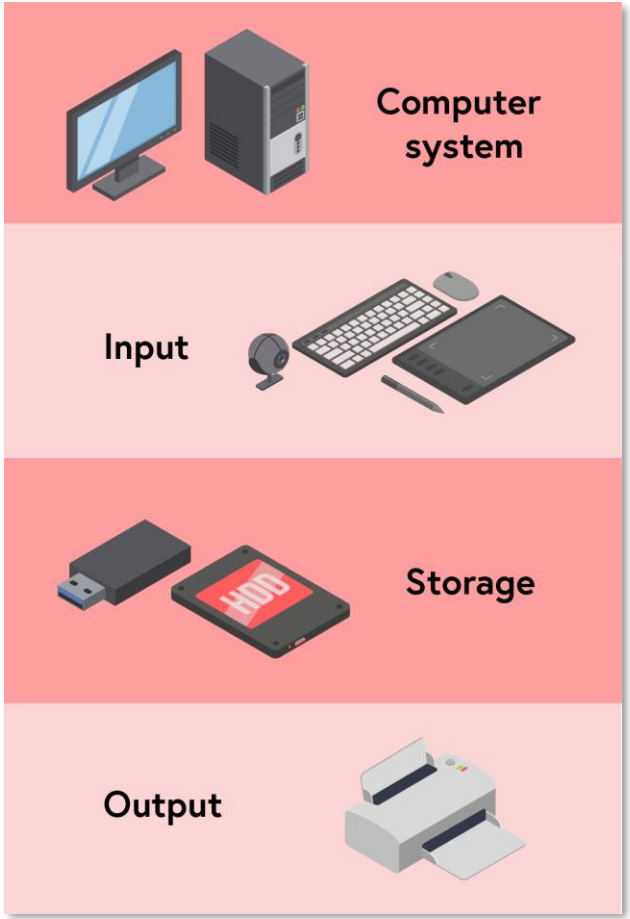
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Hardware

- tangible/physical foundation of any GIS
- **devices and machinery** required to **collect** (**GPS**), store (hard drives), process (CPU/GPUs), and display **geospatial data** (screens, printers)

What are pieces of hardware in this room that could be used in a GIS workflow?



@Future Learn

Software

- **tools and functions** necessary to manipulate, analyze, and visualize spatial data
  - Relation Database Management System (RDBMS)
  - Desktop GIS
  - Programming languages
  - Extract, transform, load (ETL)

What are modern examples?

ArcGIS Online, Google Earth Engine, Amazon Webservices



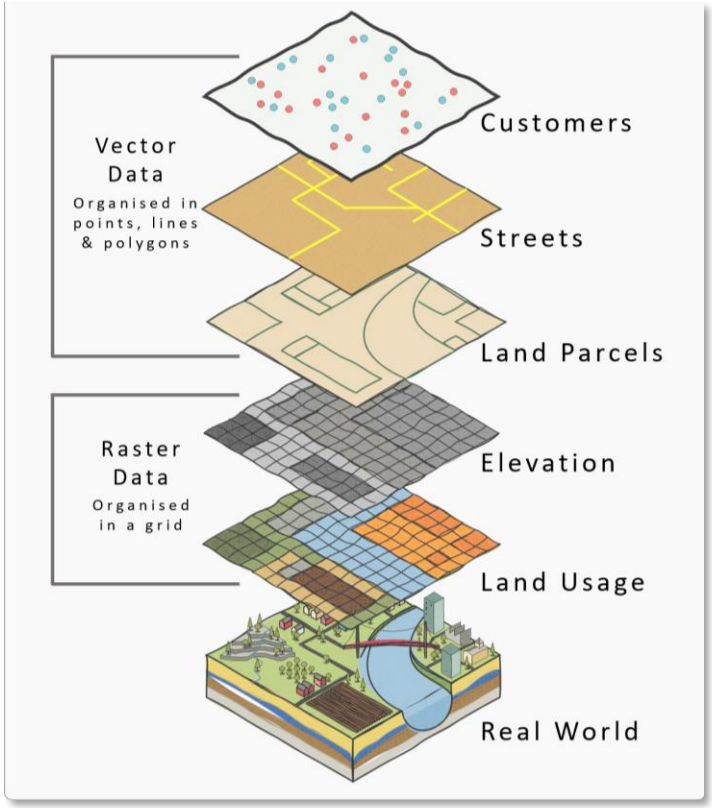
@GeographyRealm

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Data

- combination of **physical features** and it's **information**
- location, attribute, and **temporal**
- need management ensures accuracy, consistency, and interoperability across systems.

Do you have any example of geospatial data?

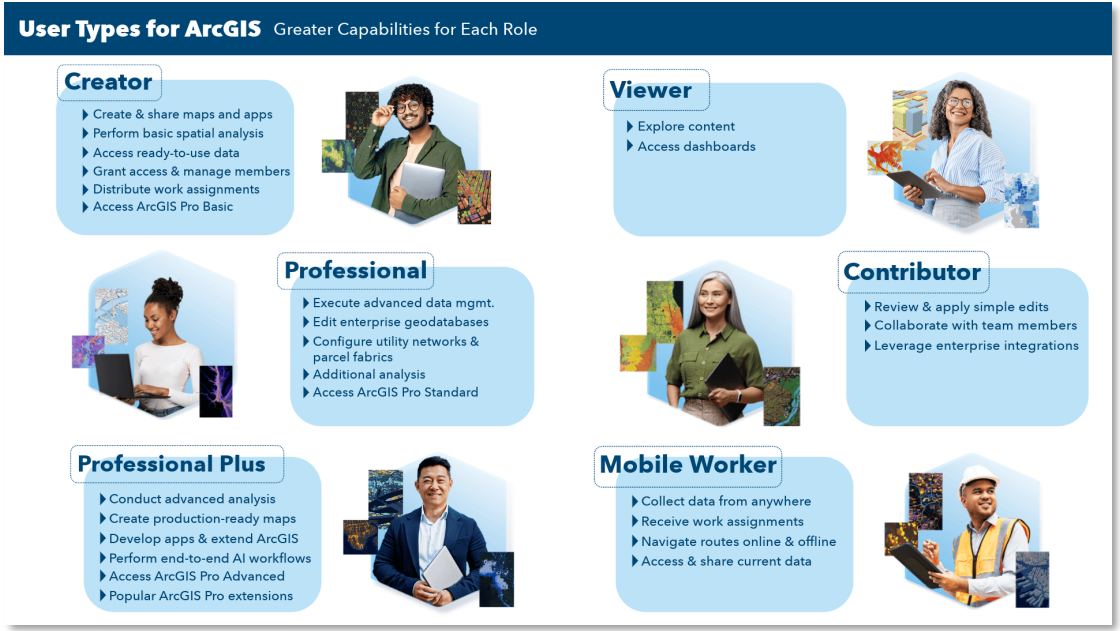


@Mapnova

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People

- **user** of the GIS system
- technical specialist
- domain experts
- decision-makers
- end users
- broad population
- **you!**

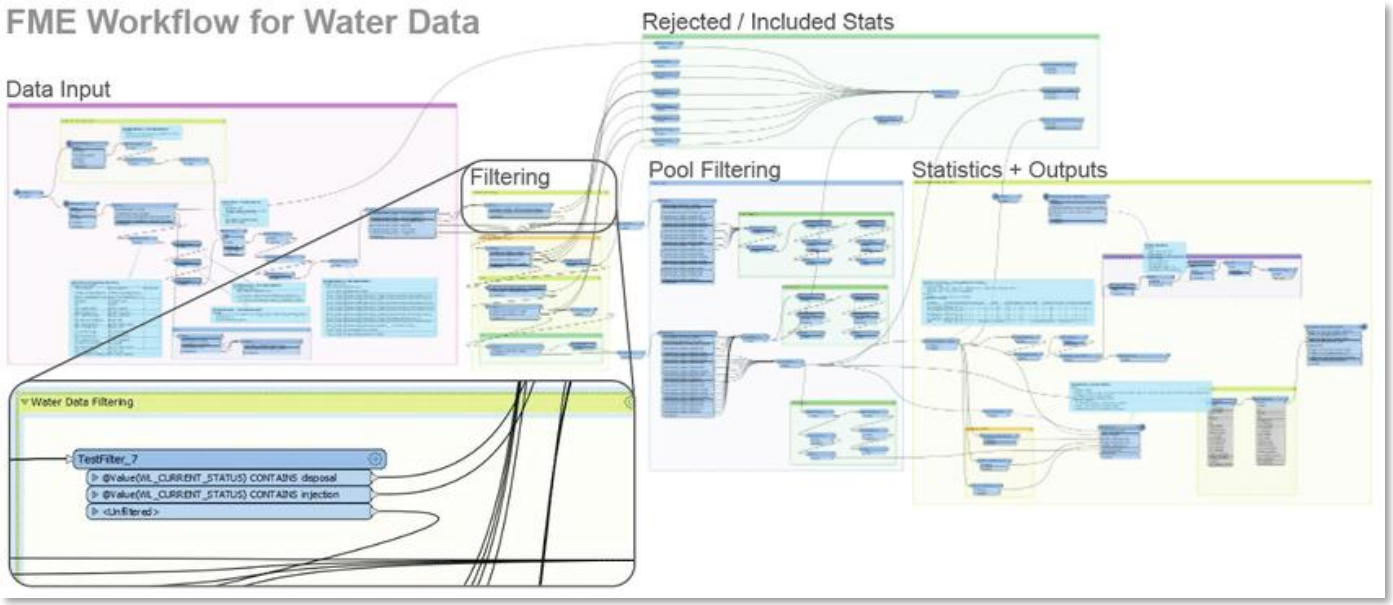


@ESRI

Who uses GIS in the real world? What are their needs and goals?

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Methods

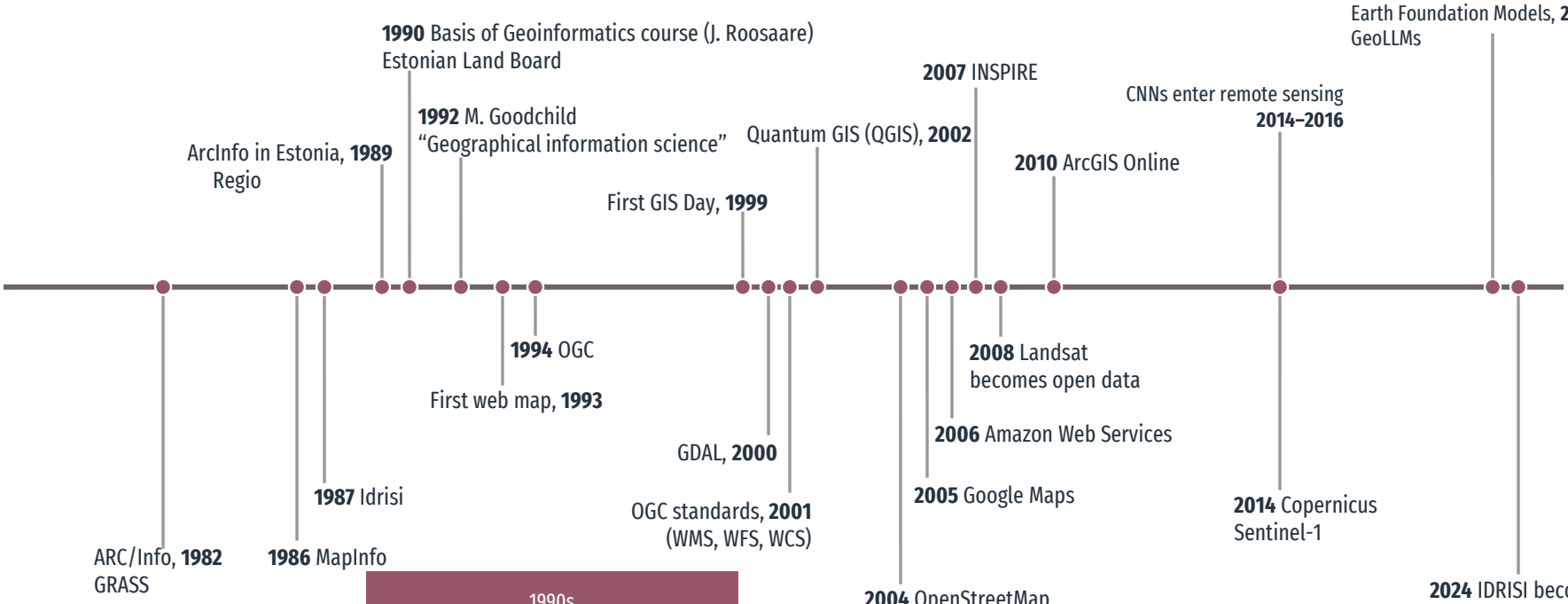


protocols, standards, and workflows that govern GIS operations, ensuring reproducibility and accuracy across applications

@Kingston, A. W., Mort, A., Deblonde, C., & Ardakani, O. H. (2021). Hydrogen sulfide (H2S) distribution in the Triassic Montney Formation of the Western Canadian Sedimentary Basin. https://doi.org/10.4095/329266

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GIS development



Before the 1980s
Development of GIS methods

1980s
GIS software development and commercialization

1990s
GIS becomes mainstream and GIScience emerges

2000s
Web GIS

From 2010
Open-source software development

From 2018 GeoAI

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Key GIS Concept: Data Layers

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GIS, a layer cake of info

A GIS can be considered a *layer cake* of info as it organizes different types of geographic data that are stacked on top of each other to create a complete picture

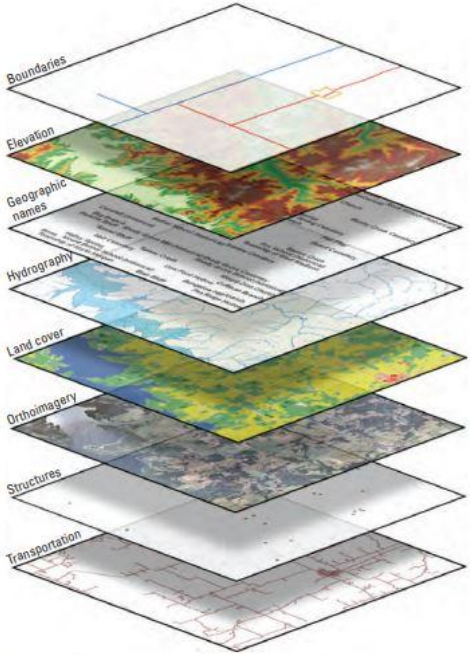


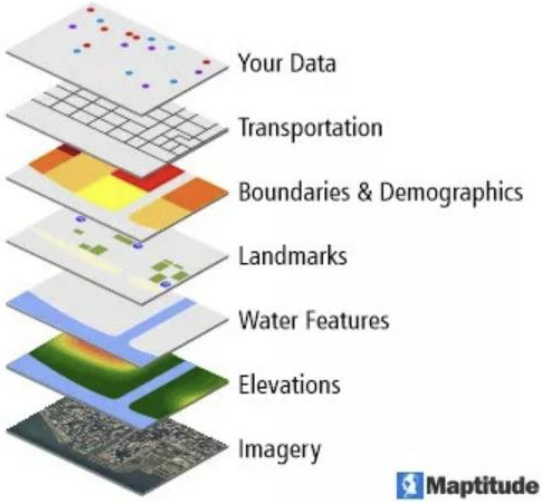
Figure 1. Eight base layers of *The National Map*.

@USGS

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Data Layers

From GIS Data Layers



To the real world

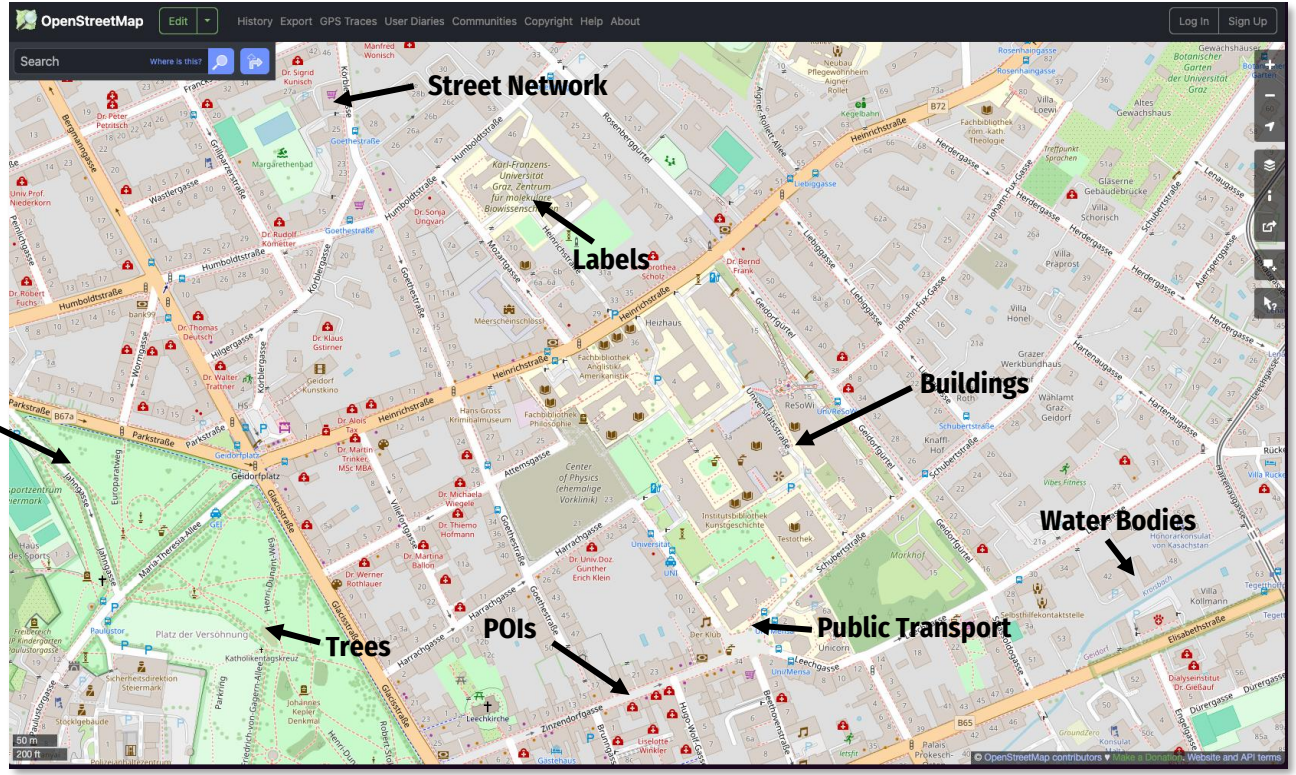


@Caliper

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Example

What data layers can you observe in this map?



© OpenStreetMap

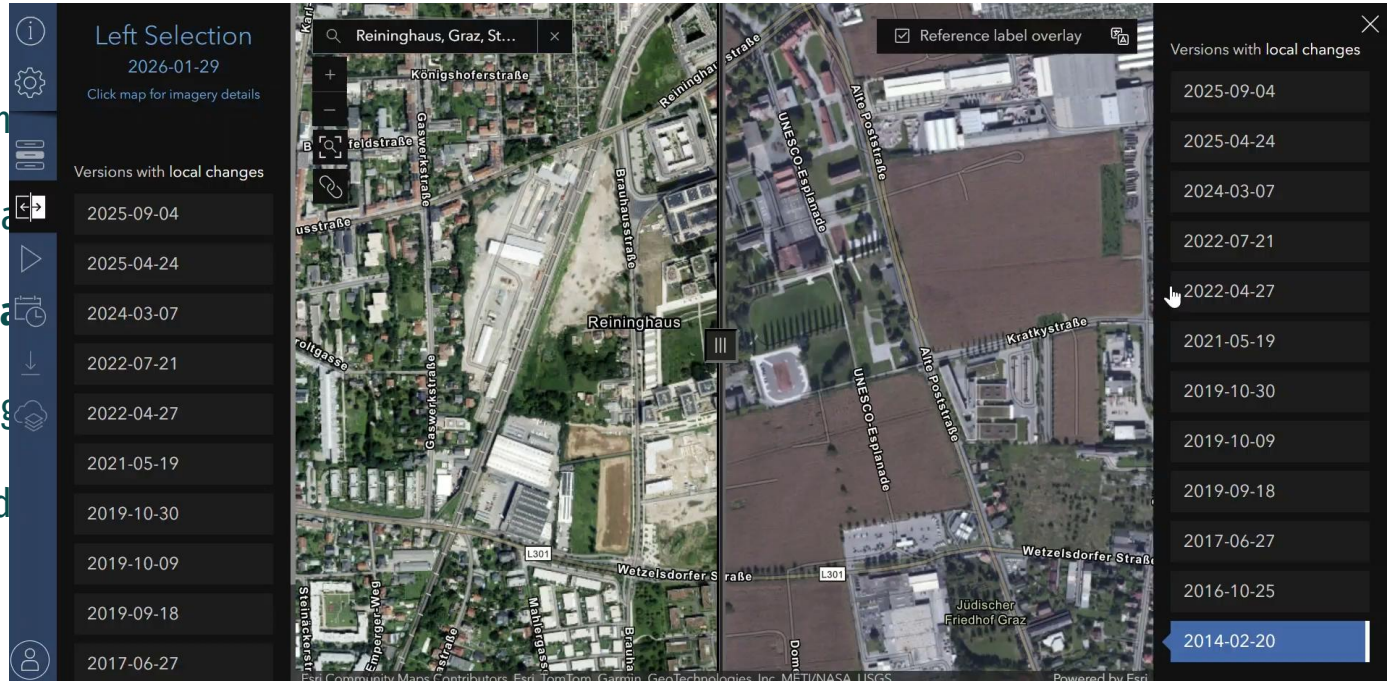
Do you think that everything is present?

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Example 2

- Location Iden
- Condition Ana
- **Temporal Ana**
- Pattern Recog
- Scenario Mod

The “Wayback“ app by ESRI



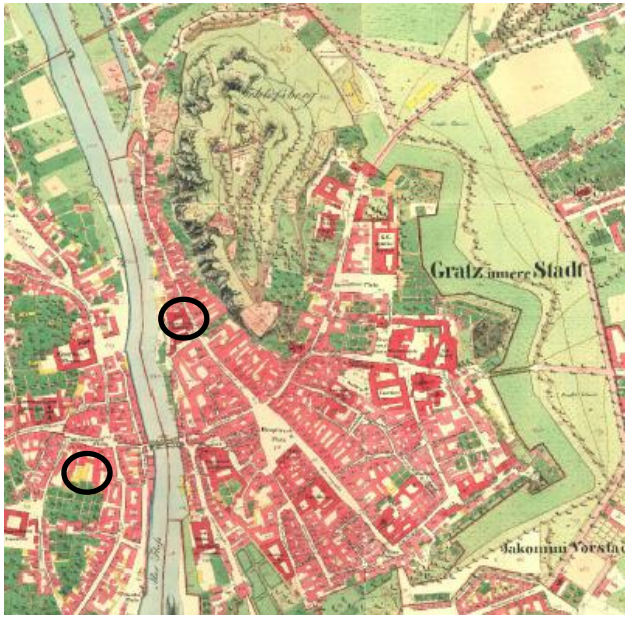
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GIS data

- Imagery
- Basemaps
- Demographic & Lifestyles
- Boundaries and Places
- Transportation
- Earth Observations
- Historical Maps



© ESA



© Das Land Steiermark

GIS applications

- Agriculture
- Water Resource Management
- Land degradation, Land use and Land cover change
- Disaster Risk Management
- Energy planning / Oil and Gas
- Climate Change Mitigation and Adaptation
- Food Security and Poverty Reduction
- Health and Telemedicine
- Land Administration / Cadastre / Urban Planning
- Socio-Economic mapping
- Utility Management
- Transport
- Media (e.g. TV: for Reporting, Marketing, Advertising, etc.)
- etc.

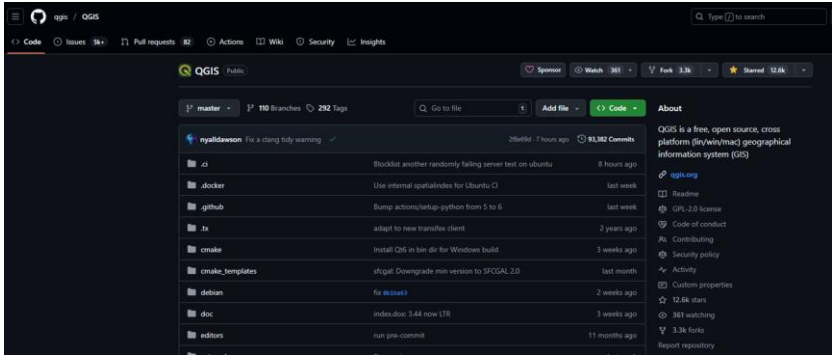


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QGIS software

Our main tool

Over 90k commits (= updates to the code of the software) and more than 500 contributors
https://github.com/qgis/QGIS.git





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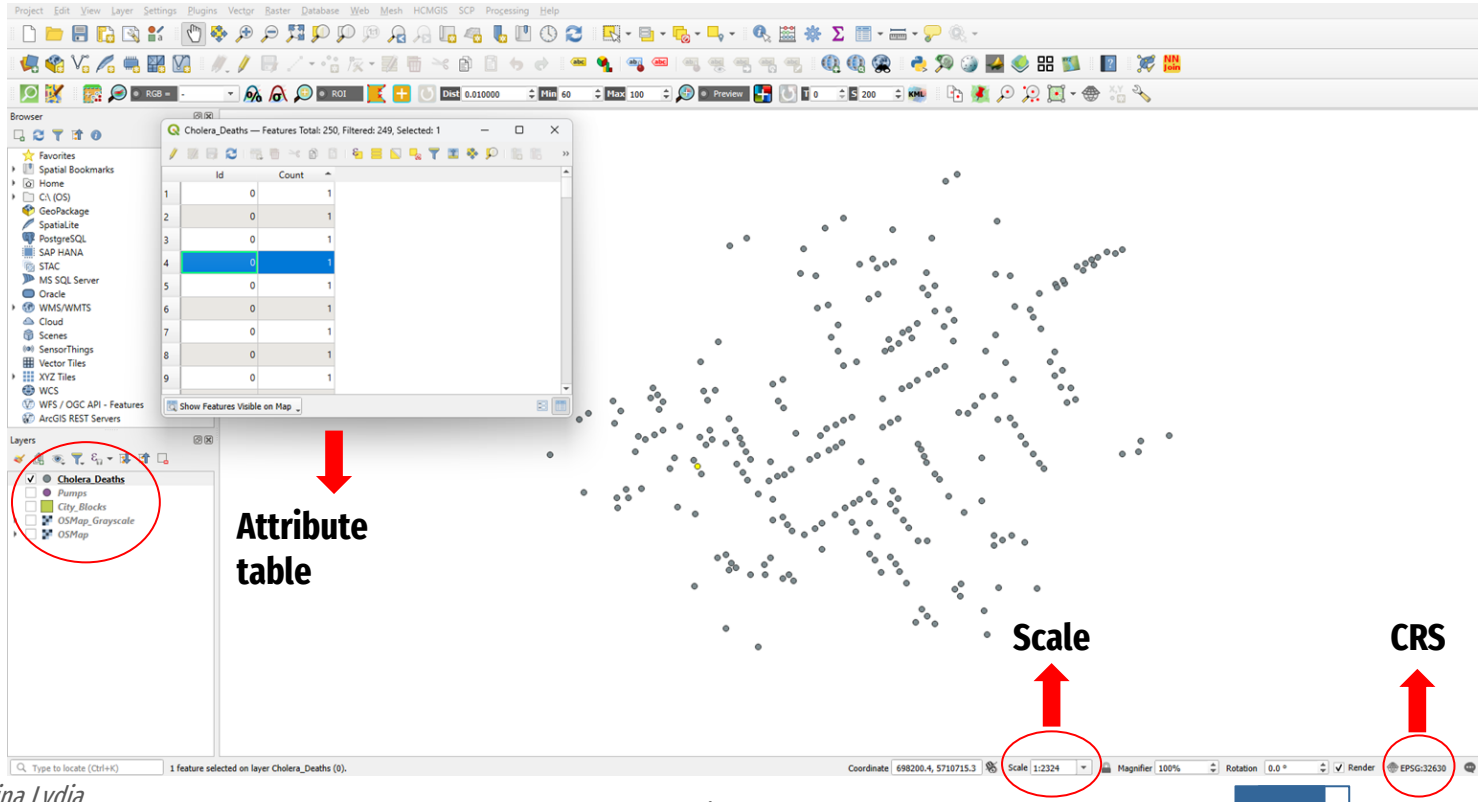
Other GIS software

GIS landscape



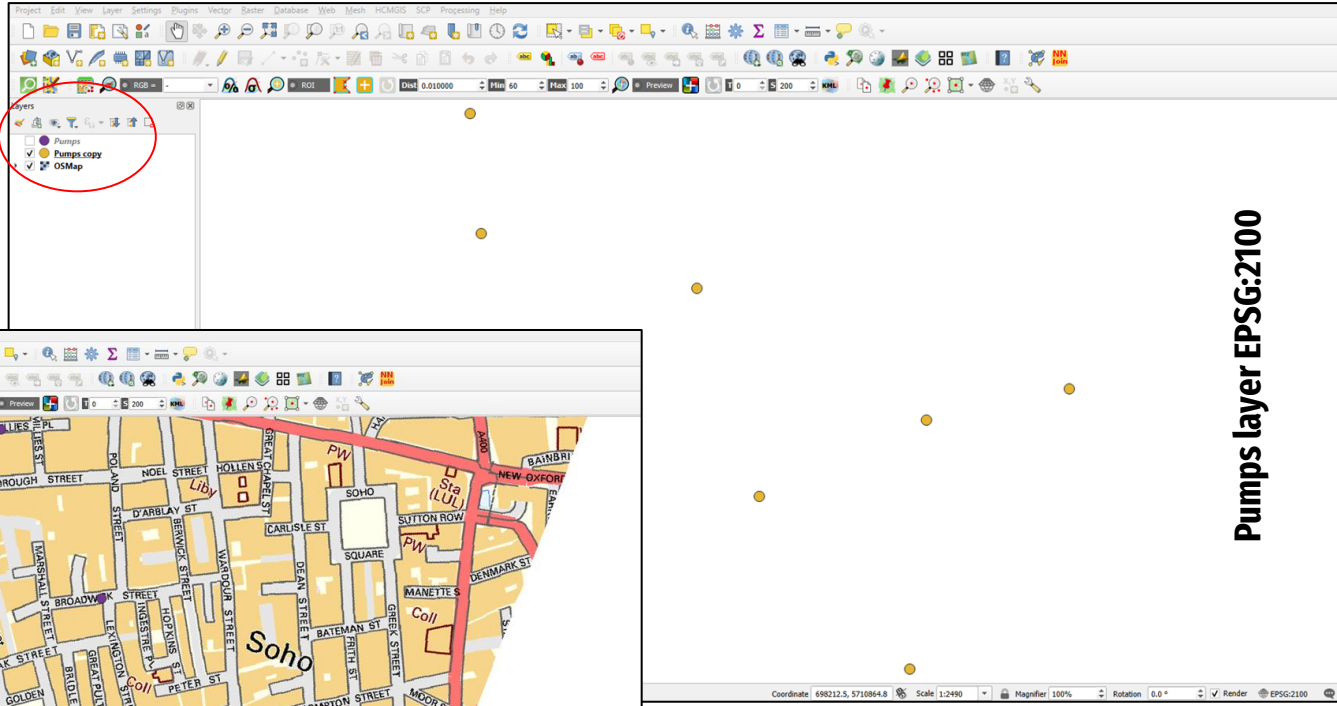
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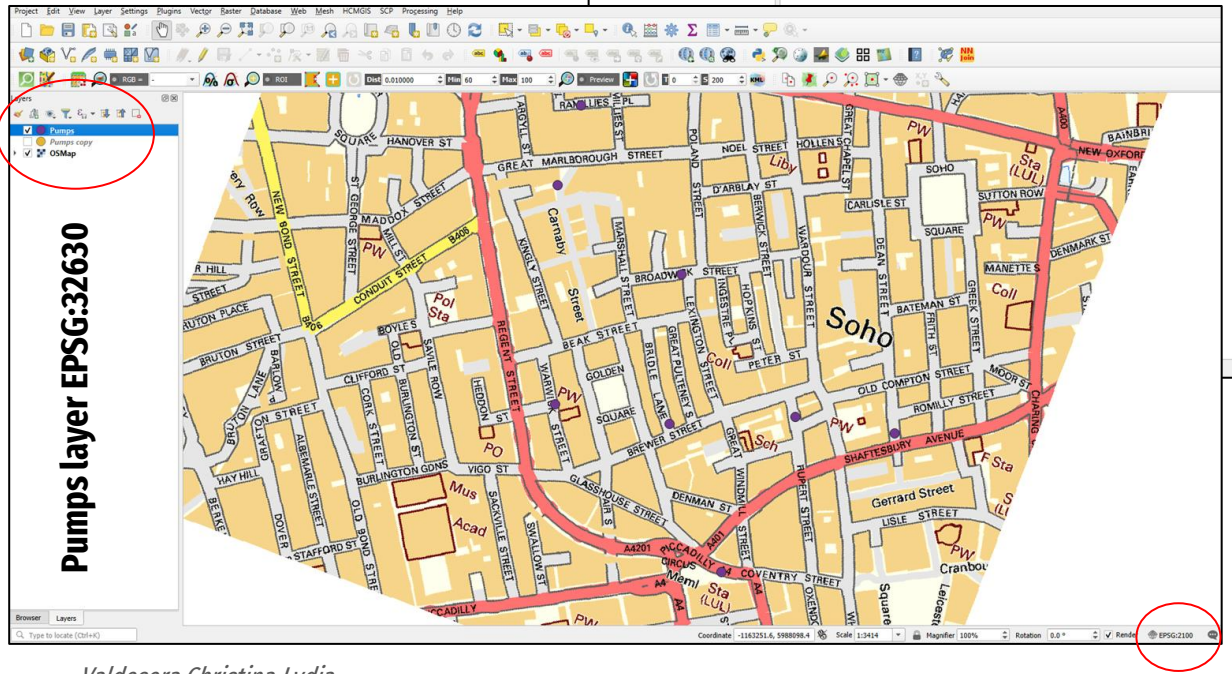
GIS elements in a snap



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CRS common pitfall





➡ **Project CRS** differs from **layer CRS**

Data structure



🤔 What is a shapefile & what are its components?

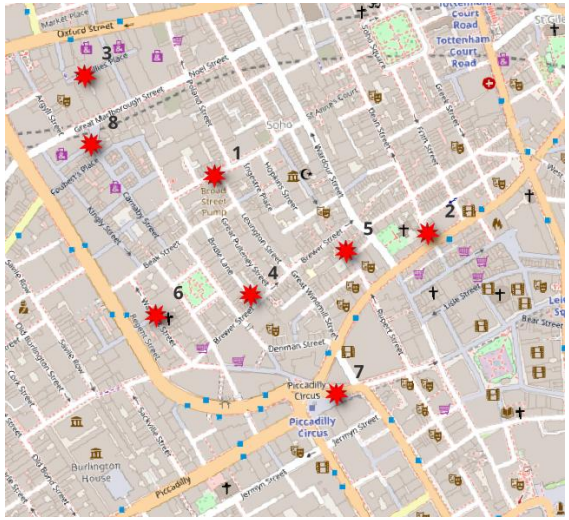
	Extension	Description
Required	.shp	The geometry of vector features are stored in this file
	.dbf	The attributes of vector features are stored in this file
	.shx	This file is an index that helps the GIS application to find features more quickly
Optional	.prj	Contains coordinate system & projection information
	.cpg	Specifies character encoding used in the .dbf file
	.sbn	Spatial index for speeding up spatial queries
	.sbx	Companion to .sbn, also for spatial indexing

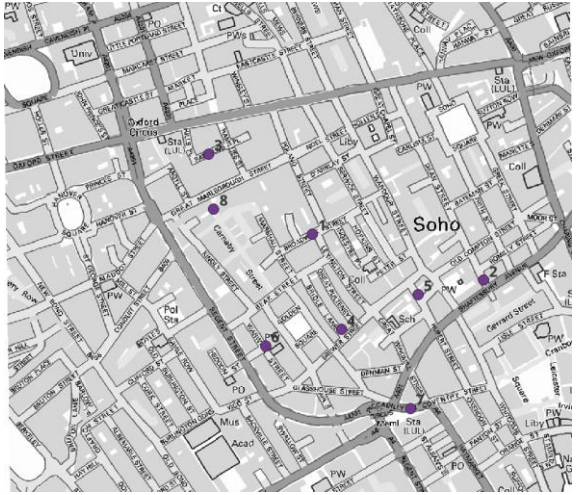
@QGIS

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Symbology

- Symbology is the selection of colors, shapes & sizes on a map
- Good symbology makes maps easy to read & understand
- It helps us quickly see patterns & differences

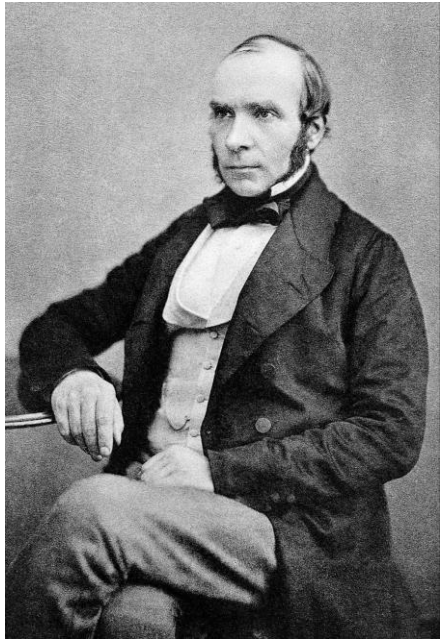






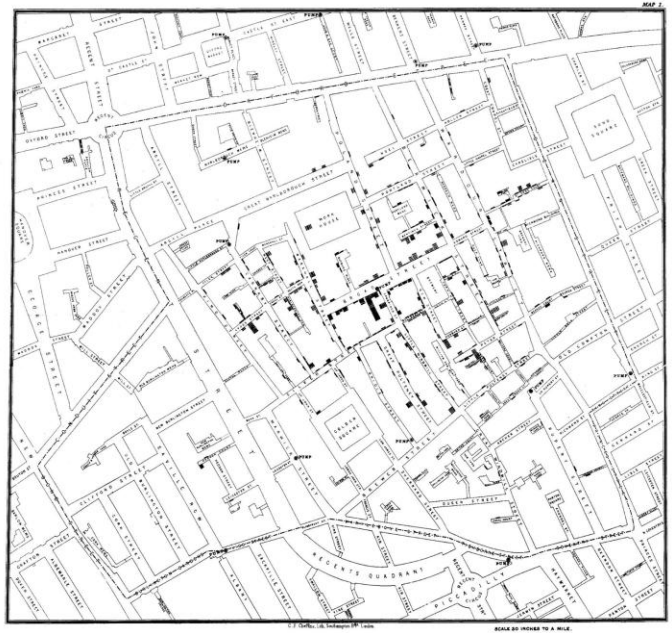
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John Snow



@Wikimedia

In the 1850s he used a primitive form of GIS analysis to identify source of cholera outbreak in Central London. He mapped the water pumps & cholera deaths



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Takeaways

- Which GIS questions have we answered with today's exercise?
- Project CRS vs layer CRS
- Which was the contaminated pump?
- Keywords to remember: geocoding, thematic mapping, choropleth map, spatial analysis